

Software Engineering LAB-3

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Given Scenario in Lab-1: Multi-Vendor Artisan E-Commerce Marketplace.

Chosen Architecture :Microservices

Architectural choice: I selected the Microservices architectural style. The Multi-Vendor Artisan Marketplace is like an app with many little apps in it which are small, independently services (Order Manager, Payment Service, Inventory Service, Vendor Payout Service, Notification Service). Each owns its own data and communicates through well-defined APIs and events.

Reason 1: Uneven load across checkout functions.

During Checkout the workload is not uniform. Festival and sale periods create spikes in order placement ,inventory updates and payment calls. Microservices let us scale only the order manager and payment service during .

Reason 2: Business rules that change independently.

The marketplace has rules that change separately like new payment methods or some law or regulations . Each rule lives in its own service, so the fee logic or a new wallet can be changed and redeployed without changing other stuff.

Security advantage: Microservices give us isolation so if one thing is taken down other things are not affected , the data is also independent c.

Example: Sensitive card and wallet handling is confined to the Payment Service, which is the only component that talks to the external Payment Gateway. Other services never see card data

Performance benefit: Independent scaling and asynchronous communication improve responsiveness. Notifications to the Shopper and each Vendor are sent through asynchronous events, so the Shopper sees the order confirmation immediately without waiting for emails or SMS